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Remote sensing identification and model-based prediction of harmful algal blooms in inland waters: Current insights and future perspectives

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ABSTRACT

Harmful algal blooms (HABs) in freshwater systems pose significant threats to water quality, ecological stability, and public health. Managing these blooms requires substantial resources, making early and accurate prediction essential. Remote sensing technologies have emerged as powerful tools for HAB identification and forecasting, providing critical data to support predictive modeling. However, forecasting HABs remains challenging due to inherent uncertainties in bloom dynamics. Recent advances in data science and computational methods have facilitated the widespread application of both data-driven (DD) and process-based (PB) models for HAB prediction. DD models, particularly machine learning techniques such as artificial neural networks (ANN), random forest (RF), and long short-term memory (LSTM), effectively capture relationships between environmental variables and bloom events from historical data, enabling accurate short-term predictions. In contrast, PB models simulate the biochemical processes driving algal growth, such as photosynthesis, nutrient uptake, and cell division, providing mechanistic insights and supporting targeted management strategies. Despite these advancements, challenges remain, including the selection of optimal input variables, model transferability across diverse water bodies, and the interpretability of complex machine learning models. Future research should focus on developing adaptive hybrid models, integrating interpretable artificial intelligence (XAI) techniques, and enhancing the synergy between remote sensing and predictive modeling. This comprehensive approach has the potential to provide robust early warning systems for HABs, contributing to sustainable freshwater management.

1. Introduction

Harmful algal blooms (HABs) in inland water systems represent a growing global environmental concern with significant ecological, economic, and public health implications. Typically triggered by eutrophication, these phenomena cause substantial damage to aquatic ecosystems worldwide. Under suitable conditions, algae multiply rapidly, creating extensive blooms that block sunlight and form dense cell clusters, which deplete dissolved oxygen and may cause hypoxia. Severe hypoxia can result in black water masses, disrupting aquatic ecosystems and endangering aquatic life. Additionally, some toxin-producing algae release harmful substances that bioaccumulate through the food chain, posing serious health risks to humans and animals (Flynn and McGillicuddy, 2018; Ho et al., 2019; Landsberg, 2002; Rousso et al., 2020).

Beyond natural eutrophication processes, climate change exacerbates HAB occurrences by altering key meteorological variables such as water temperature, stratification, and precipitation, which in turn affect nutrient transport, runoff, and phytoplankton metabolism (Dai et al., 2023; Du et al., 2023). Additionally, extreme weather events like heavy rainfall, droughts, and heatwaves can cause nutrient fluctuations, further aggravating algal blooms (Ding et al., 2024; Jiang et al., 2022; Qin et al., 2021). Recent studies indicate a global increase in cyanobacterial blooms over the past few decades, attributed to ongoing eutrophication, rising CO_2 levels, and climate warming (Gobler, 2020).

Traditional methods for assessing algal biomass, such as field sampling and laboratory analysis, are labor-intensive, time-consuming, and costly, making them impractical for large-scale monitoring. Remote sensing technology offers a promising alternative by enabling high-temporal and wide-area data collection, significantly enhancing the efficiency and scalability of HAB monitoring. Satellite-based remote sensing, in particular, provides critical insights into bloom dynamics

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through indicators such as chlorophyll-a (Chl-a) concentration and surface algal bloom area, which can be integrated with machine learning (ML) algorithms to develop predictive models. Remote sensing and predictive models are increasingly viewed as complementary components in HAB forecasting systems. Remote sensing platforms provide large-scale, frequent observations of surface water conditions (e.g., Chl-a concentration, water temperature, and turbidity), which are often used as predictive variables for algal bloom occurrence. These satellite data not only facilitate early detection of algal bloom events, but also serve as important input features for HAB prediction models, enabling real-time and spatially explicit algal bloom forecasts. In addition, advances in satellite sensors and cloud-based geospatial platforms (e.g., Google Earth Engine) have significantly improved the temporal resolution and accessibility of remote sensing data, making them suitable for dynamic model integration that captures the development of algal blooms in different aquatic systems. As a result, remote sensing has become an indispensable tool for detecting the temporal and spatial distribution of phytoplankton in water bodies, gaining widespread use in recent decades (Ho et al., 2017; Simis et al., 2005). Given the ecological and economic impacts of HABs, developing robust prediction and early warning systems is essential for proactive management. Although nutrient enrichment, particularly of nitrogen (N) and phosphorus (P), is widely recognized as the primary driver of HABs, the mechanisms linking nutrient dynamics to bloom initiation and intensity remain complex and system-specific. In many freshwater ecosystems, phosphorus has traditionally been considered the limiting nutrient, leading to management strategies that focus on reducing external P inputs. However, recent studies have demonstrated that nitrogen limitation is becoming increasingly common in eutrophic and hypereutrophic lakes, particularly in regions experiencing warming, stratification, and high denitrification rates (Brehob et al., 2024; Ding et al., 2018; Qin et al., 2020; Schindler et al., 2016; Zhou et al., 2022). In shallow polymictic lakes, frequent sediment resuspension and water column mixing promote internal phosphorus loading, enabling legacy phosphorus stored in sediments to recycle continuously and fuel algal blooms. Conversely, in stratified lakes, thermal layering creates distinct nutrient dynamics: nutrients accumulate in the hypolimnion under anaerobic conditions during stratification periods, then redistribute to surface waters during seasonal turnover events, leading to delayed bloom responses (Tammeorg et al., 2024; Yang et al., 2020a, 2020b). These processes challenge the generalizability of simple input–output models and require careful consideration of site-specific nutrient dynamics. Consequently, selecting appropriate input variables, training data, and model structures must be informed by the specific ecological and biogeochemical characteristics of the target system. Current HAB prediction models generally fall into two categories. The first category consists of process-based (PB) models, which account for both internal and external factors influencing bloom formation. These models use differential equations to describe the interactions among various influencing factors, with numerous undetermined parameters. While these models provide valuable insights into the underlying mechanisms of bloom formation, they are complex and require extensive prior knowledge and empirical data for accurate parameter determination. PB models often incorporate these variables through nutrient cycling modules, but their relative importance can vary significantly in different aquatic environments.In contrast, the second category comprises data-driven (DD) models, which focus on identifying bloom drivers and developing predictive algorithms. These models often leverage remote sensing data to establish quantitative relationships between bloom occurrence and related environmental factors. The integration of remote sensing observations with these modeling approaches has created new opportunities for timely HAB monitoring and prediction at various spatial scales. Moreover, water quality indicators such as total nitrogen (TN), total phosphorus (TP), Dissolved oxygen (DO), and pH often serve as key input variables for data-driven models, influencing bloom initiation and intensity. The predictive accuracy of models tends to vary depending on the availability and variability of these parameters across systems (Lin et al., 2023; Wherry and Schenk, 2024).

Despite significant advances in both remote sensing technology and modeling approaches, substantial challenges remain in accurately predicting HAB occurrences across diverse inland water systems. This study reviews recent advancements in HAB prediction models, with a particular focus on modeling approaches and their associated challenges. Additionally, we discuss the role of remote sensing in bloom identification and its potential integration with prediction frameworks (Fig. 1). Unlike previous reviews that tend to treat detection and prediction separately, this study bridges the gap between satellite-based bloom identification and predictive modeling, offering a unified perspective on how remotely sensed features can be utilized in different modeling frameworks. This paper is organized as follows: Section 2 examines various remote sensing indicators for monitoring algal blooms, highlighting the strengths and limitations of each option. Section 3 reviews the application of mechanistic and empirical models in algal bloom prediction, offering a comparative analysis of these modeling approaches. Section 4 identifies current challenges in bloom prediction modeling and discusses potential methods and techniques for future optimization, including the emerging role of interpretable artificial intelligence (XAI) techniques for interpreting complex machine learning models.

2. Remote sensing technologies for HAB identification

The spectral properties of water play a crucial role in the remote sensing identification of algal blooms. These properties are primarily influenced by the absorption and scattering effects of four key substances: pure water, floating algae, non-pigmented suspended solids, and colored dissolved organic matter (CDOM) (Cao et al., 2020; Xue et al., 2019). Each of these components produces a unique reflectance signature across different wavelengths. For instance, algae-free waters typically exhibit much lower reflectance in all spectral bands compared to waters affected by algal blooms, where increased algal concentration—especially in the near-infrared region—leads to higher overall reflectance. When blooms develop, they often generate a pronounced "steep slope effect" and a "near-infrared platform," phenomena that are analogous to the spectral characteristics of terrestrial vegetation (Arabi et al., 2020; Zhang et al., 2018). These distinctive spectral features underpin the development of remote sensing algorithms, which have progressively advanced from simple threshold-based indices to sophisticated machine learning frameworks—such as random forests and support vector machines—that can effectively disentangle complex optical signals in eutrophic waters. In recent years, the integration of various remote sensing technologies has enabled the establishment of a multi-level monitoring system, wherein complementary data sources work in concert to provide comprehensive and timely assessments of HABs. Drawing on these spectral characteristics, researchers worldwide have developed remote sensing techniques to identify algal blooms. Remote sensing identification of HABs can be categorized into three approaches based on data input requirements and processing complexity. These methods fall into three main categories: pixel extraction of algal blooms from remote sensing images, retrieval methods for algal pigments, and retrieval methods for estimating total algal biomass (Fig. 2). These categories represent a progression from basic spectral detection to comprehensive biomass quantification, each serving different monitoring objectives and requiring different levels of data integration. Table 1 presents the specific remote sensing data sources and HAB identification methods, classified by indicator type.

2.1. *Target objects of HABs are identified by remote sensing*

2.1.1. *Pixel extraction*

Over the past two decades, numerous methods have been proposed for identifying algal blooms in optical remote sensing imagery.

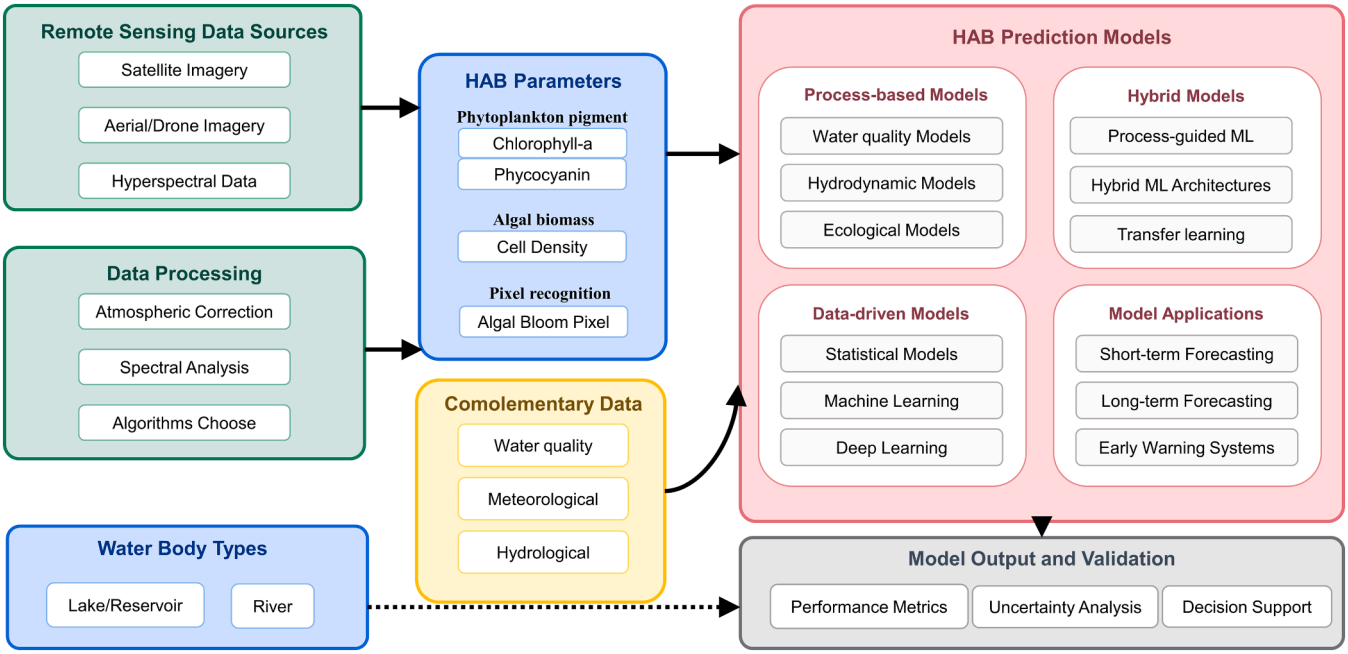

Fig. 1. Integrated framework for remote sensing identification and prediction of HABs in inland water.

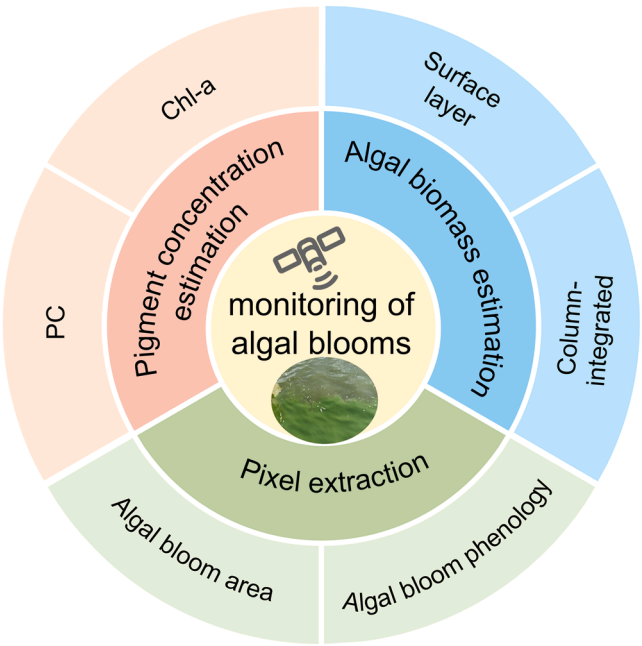

Fig. 2. Key indicators for remote sensing identification of HABs.

Generally, pixel-based extraction methods fall into two categories: index-based (or band combination threshold) methods and image-classification-based interpretation methods.

The threshold methods established mainly include single band (Ho et al., 2019; Kislik et al., 2022; Xu et al., 2021), the Normalized Difference Vegetation Index (NDVI) method (Kiage and Walker, 2009), the Floating Algal Index (FAI) (Chen et al., 2019b; Guan et al., 2023; Kiage and Walker, 2009), the Normalized Difference Chlorophyll Index (NDCI) (Kislik et al., 2022) and the Maximum Chlorophyll Index method (MCI) (Alikas et al., 2010; Binding et al., 2013) method. The index threshold extraction method has good recognition of algal blooms, but the threshold will change for remote sensing images from different data sources or even from the same data source at different times. Therefore, the selection of the threshold is highly subjective and uncertain. Furthermore, the threshold extraction method will inevitably misclassify substances with "different spectra", which can easily lead to a larger algal bloom area.

Image-classification-based interpretation methods, on the other hand, can extract characteristics from images and have also been applied in the field of remote sensing. For instance, traditional Support Vector Machines (SVM) (Zhang et al., 2020) and Artificial Neural Networks (ANN) (Grendaite and Petkevicius, 2024) have been employed to develop models for identifying algal blooms in lakes. Deep learning has also been gradually used in the identification of algal blooms. Zhu et al. (2023) built a dual U-Net model incorporating an attention gate architecture for the seasonal identification of algal blooms and achieved over 90 % accuracy and faster processing times compared to traditional methods and a single U-Net model. The advantage of supervised classification is that it can effectively utilize the experience of researchers and get rid of the trouble caused by threshold selection. However, it is easily affected by image quality problems (such as clouds, fog, etc.). It is often misunderstood during the selection of samples, which makes it difficult to obtain higher interpretation accuracy. Therefore, in essence, improving the accuracy of sample selection and considering more image information features in supervised classification are effective ways to improve the accuracy of water bloom extraction.

2.1.2. *Pigment concentration estimation*

Algal bloom identification based on pixel provides a qualitative view of algal spatial distribution, but lacks quantitative precision in determining algal concentration within the water column. Phytoplankton pigments, such as Chl-a and phycocyanin (PC), are optically active substances that function as key water quality indicators, allowing for quantitative assessment of eutrophication and algal bloom intensity. Elevated Chl-a concentrations, detectable through satellite imagery, serve as a proxy for phytoplankton biomass and indicate the presence of algal blooms (Klemas, 2012; Palacios and Mehta, 2017; Srinivasan et al., 2019).

Remote sensing retrieval of phytoplankton pigments currently relies on analytical, empirical, semi-empirical/semi-analytical, and ML algorithms trained on multispectral or hyperspectral data (Stumpf et al., 2016).The development of these algorithms follows two main modeling

Table 1
The most common applications of remote sensing-based HAB identification.

Category	Data Source	Spatial Resolution	Indicator	Method	Application	Reference
Pixel extraction	MODIS	250 m, 500 m, 1 km	Algal bloom at the pixel scale	Adjusted Floating Algae Index (FAI) index	Lake Taihu	(Jia et al., 2019)
	Sentinel-2 SMI, Landsat-8 OLI	10–60 m,30 m	Algal bloom at the pixel scale	Single band	Lake Taihu	(Xu et al., 2021)
	GOCI	500 m	Algal bloom at the pixel scale	Adjusted Floating Algae Index (AFAI) index	Lake Taihu	(Wang et al., 2023)
Pigment concentration estimation	VIIRS	750 m	Chl-a	Deep Neural Network (DNN)	217 large Chinese lakes	(Cao et al., 2024)
	Landsat-8 OLI	30 m	Chl-a	Reflectance-based approaches	Jordan Lake Reservoir	(Keith et al., 2018)
	MERIS	300 m	Chl-a, PC	Semi-analytical algorithms	Lake Balaton	(Riddick et al., 2019)
	AISA Eagle	/	Chl-a, PC	Convolutional Neural Network (CNN)	Baekje weir (located in Geum River)	(Pyo et al., 2019)
Algal biomass estimation	Sentinel-3 OLCI	300 m	Algal density	Semi-analytical algorithm	Lakes Taihu, Chaohu and Dianchi.	(Guo et al., 2024)
	MODIS/Aqua	250 m	Column-integrated algal biomass	Decision tree	Lake Chaohu	(Liu et al., 2021)
	Sentinel-3 OLCI	300m	Column-integrated algal biomass	Top-down approach combined with bivariate models	Lake Chaohu	(Yu et al., 2024)

approaches: purely statistical empirical modeling and semi-analytical modeling based on the physics of the interaction of light with the water surface.

The analytical approach is grounded in radiative transfer theory and bio-optical modeling. It derives the relationship between water quality parameters and both inherent optical properties (IOPs) and apparent optical properties (AOPs) based on the optical characteristics of water constituents (Garver and Siegel, 1997). While this method is theoretically rigorous, it involves numerous parameters and idealized assumptions, which limit its effectiveness in practical applications (Giardino et al., 2007). In contrast, empirical models establish statistical relationships between remote sensing reflectance and in situ water quality measurements by identifying spectral bands or band combinations with high correlation to specific parameters (Chang et al., 2015; Dörnhöfer and Oppelt, 2016). Traditional band ratio algorithms—such as the red-near infrared ratio (Gilerson et al., 2010) and the NDCI (Mishra and Mishra, 2012) are widely used empirical methods for Chl-a retrieval in inland water bodies. A more recent study by Sòria-Perpinyà et al. (2020) developed a dual-band ratio algorithm using Sentinel-2 imagery for PC retrieval in the Albufera of Valencia (Eastern Iberian Peninsula). Their model, based on the R740/R665 band ratio, achieved an R^2 of 0.775 and a relative RMSE of 40 %. Although simple and efficient, the model's transferability is limited, requiring local calibration before being applied to other lakes.

The development of semi-analytical algorithms marked a significant advancement in the field, as these methods derive inherent optical properties from remote sensing reflectance based on radiative transfer theory. The four-band semi-analytical model (FBAPC) has been developed to estimate PC concentration with reduced uncertainties, especially in turbid waters (Liu et al., 2018). High-resolution airborne hyperspectral imagery has demonstrated effectiveness in monitoring the spatial distribution and concentration of PC and Chl-a, providing substantial optical feature information (Pyo et al., 2018). Dev et al. (2022) developed a novel semi-analytical approach for PC and Chl-a estimation from hyperspectral data, achieving mean absolute percentage differences of 16 % for Chl-a and 28 % for PC. Their PC algorithm was derived from absorbance-concentration relationships, while the Chl-a algorithm was based on a conceptual three-band structure model. Notably, sensitivity analysis demonstrated that backscattering and other water constituents had minimal impact on PC estimation accuracy (~2 % MAPD), with reliable PC/Chl-a obtainable at PC concentrations above 10 mgm^{-3}.

The emergence of ML approaches has significantly advanced pigment concentration estimation by automatically capturing complex nonlinear relationships in spectral data. For instance, SVM have demonstrated high accuracy in estimating Chl-a concentrations in turbid waters, outperforming traditional empirical methods (Wang et al., 2018). Recent studies have further validated the effectiveness of ensemble and neural network-based models. Beal et al. (2024) applied RF and ANN models to Sentinel-2 MSI and Sentinel-3 OLCI imagery, achieving R^2 values of 0.47 (0.42) for Chl-a, 0.69 (0.22) for PC, and 0.70 (0.41) for the PC/Chl-a ratio using Sentinel-2 (Sentinel-3) data. The Boruta feature selection algorithm was employed to identify the most relevant spectral variables, enhancing model robustness. Similarly, Mamun et al. (2024) showed that the RF model outperformed k-nearest neighbor (KNN), AdaBoost, and ANN methods for Chl-a prediction, achieving R^2 values of 0.61 (RMSE = 12.51 μg/L) with Sentinel-2 MSI and 0.56 (RMSE = 16.01 μg/L) with Landsat-8 OLI imagery. These findings underscore the growing potential of ML models for reliable pigment retrieval across multiple satellite platforms and optical conditions.

However, inland waters present complex optical signals that affect the complexity and accuracy of the inversion algorithm and limit its adaptability in different environments, while machine learning methods require a large number of training data sets that can represent different optical conditions (Shi et al., 2013; Sun et al., 2012). The integration of multiple sensors and the development of adaptive algorithms that can automatically adapt to different optical conditions are key areas for future research.

2.1.3. Algal biomass estimation

Compared with the satellite-derived surface algal bloom area and Chl-a concentration, algal biomass is a more stable way to reflect water quality. Monitoring algal biomass is critical for assessing aquatic health and managing bloom events (Wu et al., 2017; Ye et al., 2014). However, the vertical distribution of phytoplankton, especially cyanobacteria, can change due to meteorological, biological and hydrological factors. Continuous daily monitoring of eutrophic lakes shows that the size of surface algal blooms can vary significantly, characterized by rapid onset, variable duration, uncertain spatial distribution, and the possibility of sudden aggregation or dissipation within days or even hours (Hu et al., 2021; Xue et al., 2023).

Traditional monitoring methods focus on surface biomass, yet the vertical migration of phytoplankton can cause significant intra-day variations in surface algal concentrations, thereby impacting the reliability of remote sensing methods that rely solely on surface measurements (Bi et al., 2019; Huang et al., 2015a; Qi et al., 2018). To achieve a

more comprehensive evaluation of bloom intensity, it is essential to account for the vertical heterogeneity of algal distribution by estimating the column-integrated algal biomass. Recent studies have attempted to address this issue by using two main approaches: one that assumes a uniform distribution of algae throughout the water column, and another that models the vertical distribution (e.g., using Gaussian, exponential, or jump-type functions) to establish an empirical relationship between surface Chl-a and total biomass.

Early efforts relied on process-informed empirical models. For instance, Liu et al. (2021) developed a decision-tree–based classification framework in Lake Chaohu, using floating algae index, surface Chl-a, and wind speed to identify profile types. Nonuniform vertical profiles were modeled using a power decay function, yielding acceptable estimation results from MODIS/Aqua imagery (bias = –19.95 %). Similarly, Li et al. (2017) integrated the Baseline Normalized Difference Bloom Index (B-NDBI) with bathymetric and hydrodynamic data to estimate total biomass over an 11-year period, showing good agreement with in situ data and improved interannual stability compared to surface-only approaches.

More recent work has introduced layered regression models and empirical stratification techniques. Yu et al. (2024) proposed a bivariate algorithm that combines surface retrieval (via MCI) with subsurface estimation based on empirical vertical relationships. Their model achieved R^2 up to 0.88 in bloom conditions, outperforming conventional surface-only estimates. Bi et al. (2019) applied a similar method using OLCI data for Lake Dianchi and revealed that surface biomass accounted for only ~7 % of total biomass under calm wind (<2 m/s), highlighting the importance of meteorological controls.

In parallel, machine learning methods have emerged as powerful alternatives for capturing vertical biomass patterns. Lai et al. (2024) developed a Random Forest (RF) and Extreme Gradient Boosting (XGBoost) framework to estimate column-integrated biomass in Lake Taihu using vertically distributed in situ samples and MODIS reflectance. Both models achieved strong predictive performance ($R^2 > 0.83$, MAPE < 25 %), demonstrating that vertical profile information can be effectively inferred from surface optical signatures using supervised learning.

Despite these advances, the rapid and complex variations in vertical algal distribution remain a significant challenge for developing universally applicable quantitative models. The core of the remote sensing estimation of algae total volume considering vertical non-uniform distribution is to accurately obtain and functionally characterize its vertical distribution type and optical characteristics by combining the living habits of phytoplankton and the influence of habitat conditions (Lee et al., 2012).

2.2. Remote sensing data preprocessing and quality control

To accurately identify HABs through remote sensing products, water resources managers need to make reasonable choices between different satellite products. As illustrated in Fig. 3 (Zhang et al., 2022), the temporal resolution of satellite data has a substantial impact on the regularity and reliability of algal bloom monitoring. To demonstrate this impact, Fig. 3(a) presents the Pearson correlation coefficients between the monthly maximum bloom areas in Zhang et al. (2022) derived from simulated datasets with varying temporal resolutions and from MODIS images, while Fig. 3(b) shows the seasonal trends in these maximum bloom areas. Despite these valuable insights, remote sensing of algal blooms still faces numerous challenges. These challenges include the small size of many inland water bodies, the complex optical properties of these waters, limited availability of high-resolution free satellite data, and a paucity of field calibration data. Robust remote sensing of algal blooms depends not only on advanced algorithms but also on meticulous data preprocessing and quality control. To enhance the practical application of remote sensing in HAB detection, future research should focus on better coordination between field sampling programs and satellite observations to improve calibration and validation. Strengthening these links will enhance the robustness of HAB detection algorithms.

3. Algal bloom prediction modeling

For predicting HABs, the objective is to leverage time series data of environmental parameters using both PB and DD model approaches. This enables more accurate forecasting of bloom events, providing managers and decision-makers with scientifically grounded insights for proactive decision-making.

3.1. Process-based models

PB models focus on simulating the biochemical processes of algal growth, such as photosynthesis, nutrient absorption, and cell division. These models use mathematical equations to describe these processes and consider the impact of environmental factors on algal growth. PB models can provide a deeper understanding of ecological mechanisms and help develop targeted management strategies (Table 2). A typical PB model consists of multiple interconnected modules: a hydrodynamic module that simulates water movement and temperature stratification; a nutrient cycling module that tracks key substances such as phosphorus and nitrogen; and a phytoplankton growth module that models algal biomass changes through equations incorporating light, temperature, and nutrient limitations. These modules exchange state variables at each simulation time step, forming a feedback-driven system that mimics the

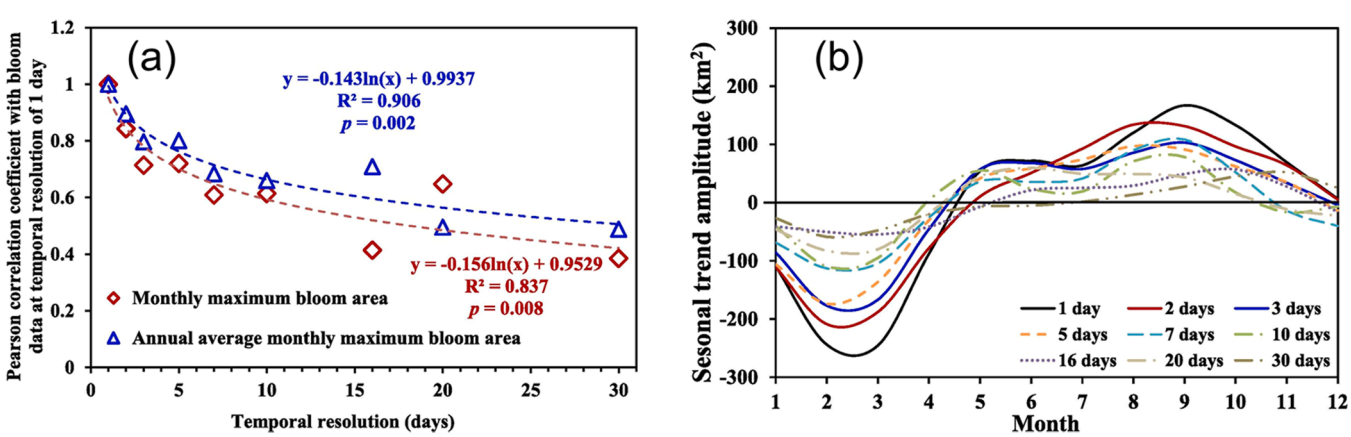

Fig. 3. The influence of temporal resolution on the regularity of algal bloom monitoring. (a) Pearson correlation coefficients between the monthly maximum areas of phytoplankton blooms in Lake Taihu between 2000 and 2020, extracted from the simulated data set with different temporal resolutions and from MODIS images. (b) Seasonal trends of monthly maximum bloom area data sets of Lake Taihu derived from different temporal resolution simulation data (Zhang et al., 2022).

Table 2
A summary of process-based models to predict HABs.

Type	Main model	Spatial dimension	Model evaluation	Reference
Water quality model	WASP	1D/2D/3D	Dynamic segmented water quality model for rivers, lakes. Includes modules for eutrophication, toxic substances, and algae dynamics.	(Zhang and Rao, 2012)
	SALMO	zero-dimension	Lake ecosystem model based on ordinary differential equations, simulating the competition and nutrient cycle of diatoms, green algae and cyanobacteria. Poor adaptability to shallow water, no consideration of wind-driven migration and recovery of benthic cyanobacteria.	(Chen et al., 2019a)
Hydrodynamic model	MyLake	2D	One-dimensional process-driven model to simulate seasonal changes in lake thermal stratification, ice sheet dynamics, phosphorus cycling, and phytoplankton biomass (e.g., chl-a) Relies on detailed meteorological data (e.g. global radiation, cloud cover) and hydrological inputs (e.g. daily streamflow, nutrient fluxes)	(Moe et al., 2016)
Ecological model	PCLake	1D	Mechanistic model for shallow lake regime shifts and nutrient thresholds. Predicts steady-state transitions under climate change and management. Requires accurate bifurcation analysis and uncertainty quantification. One-dimensional structures have difficulty capturing horizontal heterogeneity and have limited applicability to shallow lakes.	(Bucak et al., 2018)
	PROTECH	1D	A multi-species dynamic model based on phytoplankton functional groups, simulating the competition and adaptation of 18 algae such as diatoms, green algae, and cyanobacteria, suitable for community structure analysis. Species-specific parameters (such as maximum growth rate and nutrient limitation threshold) need to be calibrated, which is difficult to calibrate and has poor predictive power for cyanobacterial biomass.	(Page et al., 2018)
	WASP	1D/2D/3D	Dynamic segmented water quality model for rivers, lakes. Includes modules for eutrophication, toxic substances, and algae dynamics.	(Zhang and Rao, 2012)
	DELFT3D-BLOOM	2D/3D	Integrated with Delft3D-ECO for multi-algal competition (30+ species). Uses linear programming to optimize resource allocation among algae	(Mooij et al., 2010)
Hydrodynamic and water quality model	CE-QUAL-W2	2D	Simulates vertical stratification, nutrient cycles, and biotic groups (phytoplankton, zooplankton, macrophytes). Integrated with Tecplot for efficient visualization of long, narrow water bodies.	(Mooij et al., 2010)
	EFDC	3D	A multifunctional surface water model system including fluid dynamics, sediment transport, toxic pollutant transport, and water quality components Boundary conditions need to be defined to simulate the boundary conditions in real water bodies.	(Bae and Seo, 2018)

dynamics of real-world aquatic environments. By solving sets of differential equations, PB models provide a mechanistic, causality-based view of algal bloom development.

As a forward-derivation approach, PB models emphasize causal inference, enabling a deeper understanding of the underlying ecological mechanisms. PB models are composed of multiple modules—typically including hydrodynamics, water quality, and ecological interactions—which together facilitate a system-wide representation of algal bloom processes. The core of such models is the algal growth module,

where biomass change is expressed as the balance of growth and loss processes. Growth is usually modeled using Monod-type equations, incorporating environmental drivers such as light intensity, temperature, and nutrient concentrations (e.g., TN, TP). Loss processes include respiration, mortality, sedimentation, and predation, though often simplified due to data constraints. The evolution of PB models has followed several trajectories, with varying levels of complexity and application domains. The initial process mechanism model was a hydrodynamic model, which emphasized the influence of water flow, wind and waves, that is, the role of physical factors. Subsequently, the role of chemical factors was gradually considered, material transport and material circulation processes were introduced, and the spatial distribution of nutrients and the precipitation and resuspension of particulate matter were incorporated into the model. The biological process was further combined with the hydrodynamic and hydrochemical processes. Finally, it developed from a simplified hydrodynamic model to a comprehensive hydrodynamic-water quality-ecological coupling framework, forming a relatively complete ecological model.

For example, the Environmental Fluid Dynamics Model (EFDC) is often used to predict algal blooms in lake or reservoir areas (Wu and Xu, 2011). However, due to the complex mechanisms of algal blooms and the numerous influencing factors, its effectiveness in predicting high algal concentrations is limited (Kim et al., 2017). The Water Quality Analysis Simulation Program (WASP) model can simulate the growth and metabolic processes of aquatic organisms including cyanobacteria and predict the water quality of water bodies. In traditional WASP phytoplankton ecological dynamics models, the impact of hydrodynamic processes is primarily considered through the effects of mass migration and diffusion on algae accumulation within lake waters. However, these models do not directly account for the influence of flow regimes on algal growth. To address this limitation, Xia et al. (2022) introduced a water level regulator into the algae growth equation. This modification enhanced the algae growth module by incorporating the effects of hydrodynamic conditions on algal reproduction dynamics, thereby increasing the accuracy of simulated algal blooms. Chen et al. (2014) simulated and verified the five-year monitoring data of Meiliang Bay in Taihu Lake based on the lake water quality model SALMO, and pointed out that the increase in water temperature leads to an increase in algal biomass, and climate change poses a potential threat to the lake ecosystem.

More complex ecosystem models have been developed to capture interactions between multiple trophic levels and environmental factors. The PCLake model, developed by Janse (1997) and later extended by Mooij et al. (2010), offers a comprehensive representation of nutrient dynamics in shallow lakes, incorporating multiple trophic levels and alternative stable states. Bucak et al. (2018) linked the output of the watershed model SWAT with two different process-based lake models (PCLake and GLM-AED) and found that total phosphorus, temperature and hydraulic load were the most important variables determining cyanobacterial biomass. Future scenario simulations emphasized that reducing nutrient loads could play a key role in offsetting the effects of temperature on phytoplankton growth in the context of global warming. The PROTECH model (Phytoplankton RespOnses To Environmental CHange), developed by Reynolds et al. (2001) and refined by Elliott et al. (2010), simulates competition dynamics among multiple phytoplankton species based on their functional traits. Page et al. (2018) used the PROTECH model combined with an ensemble Kalman filter data assimilation scheme to successfully achieve short-term predictions of chlorophyll a in two mesotrophic lakes in the UK Lake District, with particularly good performance over a 4–5 day forecast period. The study also attempted to predict changes in phytoplankton community structure, but translating this into cyanobacteria predictions remained challenging due to the interchangeability of the modelled functional species. Atzori et al. (2021) developed an algal cell growth model that describes cellular processes such as photosynthesis, respiration, and nutrient uptake in detail. These models provide a more nuanced understanding of how different environmental conditions affect algal growth at the cellular level.

Hydrodynamic-ecological coupled models represent the most comprehensive framework for algal bloom simulation. The DYRESM-CAEDYM model, developed by Hamilton and Schladow (1997) and later enhanced by Hipsey et al. (2013), has been widely applied to stratified lakes and reservoirs. This one-dimensional vertical resolution model has proven effective in simulating thermal structure and cyanobacterial dynamics in systems like Lake Kinneret (Gal et al., 2009) and Lake Erie (Leon et al., 2011). For systems with significant horizontal heterogeneity, three-dimensional models such as ELCOM-CAEDYM have demonstrated superior performance. Wang et al. (2017) successfully applied this framework to Lake Taihu, capturing both spatial and temporal variability of Microcystis blooms. Similarly, Hodges et al. (2000) utilized ELCOM-CAEDYM to investigate the influence of wind-driven circulation patterns on bloom formation in shallow lakes. Li et al., 2013 Li et al. (2013) proved that the combination of three-dimensional coupled fluid dynamics-algae biomass model and algal bloom occurrence probability prediction model achieved satisfactory results in their comprehensive study of algal bloom dynamics.

Despite the progression toward increasingly sophisticated models, PB approaches face several fundamental limitations that affect their operational utility. The process mechanism model cannot expect to improve performance by continuously increasing the complexity of the model. Parameter uncertainty remains a major limitation, as highlighted by Arhonditsis et al. (2006) in their review of 153 aquatic biogeochemical models, finding that fewer than 30 % reported goodness-of-fit statistics for state variables related to phytoplankton. Robson (2014) identified structural uncertainty as another critical issue, where model formulations may not adequately represent all relevant processes. In addition, due to the limitations of environmental monitoring data, many natural process parameters cannot be observed and obtained. In actual application, some structures of the model need to be adjusted or simplified, resulting in limited prediction performance. The application of PB models in remote sensing contexts presents additional challenges. These models typically require extensive in-situ measurements for parameterization and validation, which may not always align with the spatial and temporal scales of satellite observations. To address these limitations, DD models and ensemble modeling approaches have gained popularity.

Therefore, while PB models are powerful tools for mechanism interpretation and long-term scenario analysis, their predictive utility in operational bloom forecasting is constrained without integrated high-frequency monitoring systems.

3.2. Data-driven models

The DD approach analyzes historical data sets collected from the system and uses deep learning or machine learning to discover patterns hidden in the data sets. By analyzing the data and assessing failures, machine learning techniques can build a complete cyanobacteria bloom prediction system (Zhou et al., 2016).The main model, input variables and target variable used for the algal bloom prediction models were summarized in Table 3. The internal operating process of DD models differs fundamentally from that of process-based models. Rather than simulating physical or biological processes through explicit equations, data-driven models, such as RF, SVM, Artificial Neural Networks (ANN), and Long Short-Term Memory (LSTM), learn hidden patterns directly from historical datasets. The standard workflow includes: (1) feature selection and normalization; (2) model training using labeled data to minimize prediction errors; and (3) testing or application to unseen data. Each model type has its unique architecture—for instance, RF consists of an ensemble of decision trees; ANN relies on interconnected neurons with adjustable weights; and LSTM uses memory cells and gating mechanisms to capture temporal dependencies. These internal learning processes allow data-driven models to adapt flexibly to

Table 3
A summary of data-driven models to predict HABs.

Type	Main model	Input variables	Target variables	Performance evaluation	Water type	Reference
Traditional statistical models	Multiple Linear Regression	Physical, chemical, and biological variables,	Chl-a	$R^2 = 0.973$	Lake Erie	(Mermer and Demir, 2025)
	ARIMA	Chl-a	Chl-a	RMSE = 28.98, IoA = 0.86	Taihu Lake	(Chen et al., 2015)
Machine learning models	RF	Physicochemical, hydrological, meteorological data	Total phytoplankton biomass and individual classcomposition	Total biomass simulation : $R^2 > 0.7$	Lake Mjosa	(Liu et al., 2023b)
	SVM	Water quality monitoring data, Meteorological monitoring	Chl-a	$R^2 = 0.85$, RMSE = 2.3	Pengxi River basin	(Xia and Zeng, 2022)
	XGBoost	Physical and Chemical data	Chl-a	$R^2 = 0.85$, MAE = 4.92, MAPE = 0.423, RMSE = 047	Lake Erie	(Demiray et al., 2025)
	LightGBM	Cyanobacterial and meteorological data	Bloom area	Best prediction accuracy: R = 0.36, RMSE = 0.11	Lake Taihu	(Wang et al., 2023)
Deep learning models	Bayesian networks	MODIS Terra & Aqua surface reflectance, Meteorological data, ATBI: Anti-Turbid Bloom Index (ATBI) derived from MODIS, normalized and discretized	HABs maps (presence/absence), HABs outbreak area (km^2) and proportion relative to lake area, Zonal HABs status (classified by lake regions)	Pixel-level: Acc = 0.83 (historical data), 0.78 (weather forecast data), HABs area level: Acc = 0.64 (historical), 0.62 (forecast), lake zoning level: Acc = 0.74 (historical), 0.57 (forecast)	Lake Chaohu; Lake Taihu; Lake Hulunhu	(Du et al., 2024)
	ANN	Water quality data, Meteorological data, Hydrological data	M. aeruginosaand and Chl-a	ANN_Downstream model: NSE = 0.584, $R^2 = 0.610$, RMSE = 1.50	Lake Poyang	(Huang et al., 2015b)
	CNN	Water quality variables, weather variables	Chl-a	$R^2 = 0.934$, RMSE = 5.463	Four major Korean rivers	(Lee et al., 2022)
	LSTM	Daily observation of environmental data	Chl-a	RMSE = 18.40, MAE = 13.56, $R^2 = 0.63$	Dianchi Lake	(Yu et al., 2020)
	FSN; GRU; LSTM; RF	Environmental factors , Hydrological factors , Meteorological factors , Wastewater treatment plant data	Cyanobacterial cell concentrations	FSN peformances best. $R^2 = 0.64$-0.71, RMSE = 2.06-2.26, MAE= 1.38-1.62	Nakdong River	(Shin and Cha, 2025)
	Bloomformer-2 (Based on standard Transformer)	Vertical Chl-a distribution, Seasonal hydro-meteorological Spectral clustering for layer-specific analysis	Chl-a	Single-step prediction: MAE = 0.175–0.394, MSE = 0.042–0.305, MAPE = 0.228–2.279; Multistep prediction: MAE = 0.184–0.505, MSE = 0.101–0.378, MAPE = 0.243–4.011	Taihu Lake	(Qian et al., 2024)
Hybrid models (PB+DD)	EFDC-CNN	Observed field monitoring data: Cyanobacterial cell concentrations; Chl-a distribution maps from hyperspectral image sensing; Simulated water quality outputs from hydrodynamic model	Cyanobacterial cell concentrations	RSE = 0.86; RMSE = 0.65	Hapcheon-Changnyeong Weir	(Pyo et al., 2021)
(DD+DD)	TAB	Hourly meteorological data, environmental variables (Chl-a included)	Chl-a	$R^2 = 0.74$, MAPE = 21.22%, RMSE = 4.12, KGE = 0.78	Jiulong River	(Zhang et al., 2025)

IoA: Index of agreement; RMSE: Root Mean Square Error; MSE: Mean Squared Error; RSE: Relative Standard Error; MAE: Mean Absolute Error; MAPE: Mean Absolute Percentage Error ; Acc: Accuracy.
TAB: (a Temporal Convolutional Network (TCN), an attention mechanism and Bidirectional Long Short-Term Memory (BiLSTM) network were integrated).

complex, nonlinear relationships within environmental systems.

3.2.1. *Traditional statistical models*

Traditional statistical models have been widely applied in algal blooms prediction, primarily by establishing linear relationships between environmental variables and algal growth. These models, such as simple linear regression and multiple linear regression, have been instrumental in forecasting the occurrence and progression of algal blooms.

The univariate linear regression approach represents the most basic form of statistical modeling. For instance, in one study, a linear regression model was developed between fluorescence intensity and algal concentration using laboratory-measured excitation fluorescence spectra. This model not only predicted algal concentration but also enabled the rapid identification of dominant algal populations (Shan et al., 2021). Multivariate linear regression techniques have also been extensively utilized, demonstrating their broad applicability. Illustratively, Wang et al. (2016) examined the combined effects of multiple environmental variables on algal bloom thresholds. Additionally, to address time-series data, models like the Autoregressive Integrated Moving Average (ARIMA) were developed to forecast chlorophyll-a concentrations, showing superior performance compared to multivariate regression models (Chen et al., 2015). Although ARIMA is widely used in many application areas, it can only model linear patterns in time series and cannot capture nonlinear patterns present in many complex systems. In addition, the ARIMA model relies entirely on past measurements (the variable to be predicted) and does not apply to multivariate time series.

Despite the limitations of traditional models due to their linear assumptions, they hold significant value in algal bloom prediction by providing a foundational understanding of the influence of environmental factors on algal growth. When integrated with modern data-driven techniques, these traditional linear models can still play a critical role in ecological forecasting.

3.2.2. Machine learning models

Compared with traditional statistical models, ML, especially DL models, has shown stronger predictive capabilities and improved the accuracy and timeliness of algal bloom predictions (Vinçon-Leite and Casenave, 2019). Advanced ML algorithms such as SVM, Gradient Boosting Decision Tree (GBDT), and RF have become important tools for predicting algal blooms in recent years and have achieved considerable success. These models use historical environmental data to learn and identify patterns before algal bloom events and can create adaptive and predictive models. These datasets are divided into training, testing, and validating the actual model.

Bayesian networks. Bayesian networks (BN) are a probabilistic modeling method that captures linear, nonlinear, combinatorial, and stochastic predictive relationships among multiple variables (Yu et al., 2004). They can effectively combine different data sources and handle data uncertainty through probabilistic inference and are increasingly valued in algal bloom prediction.

Cha et al. (2014) applied BN mode to Lake Paldang, South Korea, assessing the HABs risk based on multiple environmental condition thresholds. These risk levels were then translated into alert levels to aid water managers in decision-making. Similarly, Bertani et al. (2016) developed a BN model for Western Lake Erie, Canada, to evaluate various phosphorus reduction scenarios and their potential effects on the occurrence of HABs.

More recently, Liu et al. (2023a) proposed a Bayesian scale-mixture of skew-normal (SMSN) model to tackle the challenges associated with highly skewed and over-dispersed algal biomass data, facilitating the prediction of algal blooms using readily available water quality parameters such as dissolved oxygen and pH. The SMSN model accurately predicted biomass variations with skewed distributions in river and lake systems, within-sample and out-of-sample R^2 values ranging from 0.533 to 0.706 and 0.412 to 0.742, respectively. Additionally, the Bayesian framework enabled a probabilistic assessment of algal blooms, achieving an accuracy above 0.77 and a macro F_1 score exceeding 0.72, significantly reducing the inaccuracies associated with conventional point-prediction models by up to 34 %. This study demonstrated the potential of the Bayesian SMSN model for real-time prediction of algal biomass fluctuations and in-situ probabilistic assessments of algal bloom events. Du et al. (2024) applied Bayesian theorem to predict HABs in large eutrophic lakes in China for 1, 4, and 7 days in advance. The model demonstrated high accuracy, achieving a pixel-level accuracy of 0.83, a regional prediction accuracy of 0.74, and an accuracy of 0.64 for predicting the extent of algal bloom areas in the lakes. Haakonsson et al. (2024) applied a Bayesian model to estimate threshold levels of cyanobacterial biomass (CB) as indicators of bloom risk in the Río de la Plata estuary. These thresholds were used to identify favorable environmental conditions for bloom occurrence at different time scales (1 day, 7 days, and 14 days).

Support Vector Machine. Support Vector Machine (SVM) is a supervised ML technique, unlike principal component analysis (PCA), and clustering. When it is used on prior datasets, it can perform regression as well as classification. The primary process of SVM involves mapping samples to a high-dimensional feature space through a nonlinear transformation. The algorithm aims to find the optimal regression model by minimizing structural risk, thus ensuring a global solution.

Ribeiro and Torgo (2008) developed several prediction models for algal blooms in the Douro River, Porto, Portugal, using physicochemical and microbiological parameters. Their findings indicated that the SVM model outperformed the others. To further enhance the performance of SVM in algal bloom prediction, various scholars have explored different optimization strategies aimed at improving both prediction accuracy and duration. For instance, one study optimized the SVM model by incorporating rough set theory to identify key factors influencing algal bloom outbreaks and proposed a least squares support vector machine (LSSVM) for prediction. This model demonstrated higher accuracy and longer forecasting periods for medium-term algal bloom predictions compared to artificial neural networks (Liu et al., 2009). Similarly, Xia and Zeng (2022) used water quality and meteorological data to predict Chl-a concentrations with an optimized SVM model. By applying the Grey Wolf Optimizer (GWO) algorithm, they further enhanced prediction accuracy, enabling high-precision chlorophyll-a forecasting and early warning of algal blooms in the study area.

Random Forest. The Random Forest (RF) algorithm is composed of multiple decision tree models working in an ensemble. In a decision tree, input data is split into two groups based on specified criteria, with this process repeating until a stopping condition is reached. A key drawback of decision trees is their tendency to overfit the training data, resulting in poor generalization to new datasets. RF models overcome this limitation by using only a random subset of the input data to construct numerous individual trees. Each tree is built using a different random subsample of the data, and the final prediction is obtained by aggregating the outputs from all the trees in the forest (Schonlau and Zou, 2020; Yajima and Derot, 2017).

Yajima and Derot (2017) implemented a RF model with a sliding window strategy to forecast chlorophyll-a concentrations in the saline waters of Japan's Urayama Reservoir and Lake Shinji. Their findings revealed that the RF model effectively predicted the Chl-a time series trends in both water bodies. Additionally, the study highlighted that the optimal leaf number and combination of pre-selected predictors differed between lakes and reservoirs. The most influential parameters for the RF model in both environments were identified as biochemical oxygen demand (BOD), chemical oxygen demand (COD), pH, andTN/TP. One study demonstrates that the RF model outperforms classical machine learning approaches, such as SVM and neural networks (NN), in predicting Pseudo-nitzschia spp. blooms in the Galician Rias Baixas. The RF model shows a better balance between sensitivity and specificity, indicating its greater adaptability to unbalanced datasets and improved performance in bloom forecasting (Alaez et al., 2021).

3.2.3. Deep learning models

Artificial neural networks. Artificial neural networks (ANN) has emerged as powerful tool for predicting HABs in inland water bodies due to their ability to handle complex, non-linear relationships among environmental factors. The typical ANN architecture consists of input, hidden, and output layers. The input layer receives environmental parameters (water temperature, pH, dissolved oxygen, nutrients), which are processed through interconnected neurons in the hidden layer(s), with the output layer providing predicted algal bloom concentrations or probabilities. During training, connection weights between neurons are adjusted to minimize prediction errors, allowing ANN to learn from historical data and improve forecasting accuracy without explicit programming of the underlying model structure (Moreira et al., 2022).

Several studies demonstrate ANN effectiveness in this domain. Bui et al. (2017) successfully predicted cyanobacterial blooms in Vietnam's Dau Tieng Reservoir using an ANN model with eight environmental parameters as inputs, achieving low RMSE and high correlation coefficients. Similarly, Tian et al. (2017) optimized an ANN model for predicting Chl-a dynamics in a Chinese estuary reservoir by using the change in Chl-a value as the output variable rather than the base value, significantly improving predictive performance compared to traditional ANN approaches.

However, ANN also has limitations because it requires a large amount of training data and it is difficult to adjust the structural parameters which are mainly based on experience (Xie et al., 2012). Mamun et al. (2020) found that SVM model outperformed ANN when

predicting Chl-a across different areas and seasons of a reservoir, highlighting the importance of model selection based on specific application contexts.

Long Short-Term Memory. The Long Short-Term Memory (LSTM) model has gained popularity in predicting algal blooms due to its strong performance in time series data forecasting (Lee and Lee, 2018). As an advanced recurrent neural network, LSTM excels at learning long-term dependencies through its specialized memory cell design, which allows it to retain significant features of input variables while discarding less relevant information. This capability makes it particularly suitable for long-term time series modeling (Shin et al., 2019).

However, similar to other ML models, the performance of the model is significantly affected by the characteristics of the input data. Therefore, there are various studies aimed at improving the performance of LSTM by enhancing the composition of the input data (Park et al., 2024). For example, Yu et al. (2020) utilized a combined Wavelet Domain Threshold Denoising (WDTD) technique with an LSTM model to simulate historical variations in Chl-a concentrations in Dianchi Lake and forecast future trends. Their model exhibited strong predictive performance, characterized by low error rates and high generalization capabilities. More recently, Liu et al. (2022) developed a novel algae prediction method, termed WLSTM, by integrating wavelet analysis techniques with a deep learning model (LSTM). This approach demonstrated superior performance compared to state-of-the-art time series forecasting tools, including LSTM, DNN, and ARIMA, accurately predicting algal dynamics at hourly, daily, and monthly resolutions. Together, these studies highlight the effectiveness of LSTM and its hybrids in enhancing algal bloom prediction accuracy.

Transformer. The Transformer architecture has emerged as a foundational framework for developing many of the world's most advanced models, achieving significant success across various domains. In the context of water quality management, Transformer models have been integrated into broader frameworks to enhance real-time monitoring and decision-making processes. For instance, Qian et al. (2024) utilized deep learning and big data technologies to develop the Bloomformer-2 model, which enables both single-step and multi-step predictions of Chl-a concentrations. This capability is instrumental in providing early warnings for algal blooms in Taihu Lake, highlighting the potential of Transformer-based models in addressing environmental challenges. Building on this approach, Liu et al. (2025) implemented iTransformer, another model based on the Transformer architecture, which demonstrated high accuracy in predicting HABs in Dongting Lake, with a mean square error (MSE) of only 0.139. Through attention weight analysis, the iTransformer successfully identified nitrogen, phosphorus, and temperature as key factors affecting algal bloom density. These findings not only verify the model's interpretability but also provide a scientific foundation for environmental management strategies. Together, these applications highlight the potential of Transformer-based models in addressing complex environmental challenges through advanced predictive capabilities.

However, solely relying on Transformer models does present some limitations. One notable drawback is their computational complexity, as Transformers typically require significant amounts of data and processing power, particularly in the training phase. Additionally, Transformer may struggle with capturing the temporal dynamics inherent in time series data without supplementary mechanisms, as they were originally designed for static data contexts. Recent advances in Transformer-based architectures have addressed several limitations of earlier models in time-series forecasting. For example, MultiPatchFormer (Naghashi et al., 2025) introduces a multi-scale patch-wise temporal encoding mechanism combined with channel-wise representation learning, enabling the model to capture both diverse temporal granularities and intricate inter-series dependencies. Similarly, the FLUID-GPT framework (Yang et al., 2023), a generative pre-trained Transformer, enhances forecasting capabilities by incorporating domain-specific pretraining and improved adaptability to complex temporal dynamics. These developments extend the applicability of Transformer variants beyond conventional neural networks, offering improved robustness and generalization in multivariate time series prediction tasks. Notably, such capabilities are particularly valuable for ecological forecasting challenges such as harmful algal bloom prediction, where multi-scale temporal patterns and cross-variable interactions are common and difficult to model with traditional approaches. As a result, integrating Transformer models with other techniques, such as recurrent architectures or hybrid approaches, can enhance their performance and adaptability in dynamic environments, making them more suitable for tasks like algal bloom prediction where temporal patterns are crucial.

3.3. Hybrid models

Purely data-driven methods often fall short in addressing the highly variable, non-stationary nature of algal dynamics (Xiao et al., 2017). In response, recent research has turned towards hybrid modeling, which combines mechanistic and data-driven approaches, or incorporates multiple models to enhance predictive accuracy and adaptability for capturing complex algal behaviors. In this review, hybrid models are categorized into two main types: (1) hybrid PB–DD models, which integrate PB and DD approaches, and (2) hybrid DD–DD models, which combine multiple data-driven algorithms to leverage their complementary strengths.

For instance, Couture et al. (2018) integrated PB models (including a climate model, hydrological model, and MyLake) with a BN to evaluate the probability of HABs occurrences under future scenarios in Lake Vansjø, Norway. Similarly, Zheng et al. (2021) employed the EFDC to extrapolate single-point data across the entire watershed, followed by the use of LSTM to predict Chl-a for 1 to 3 time steps throughout the water column. The combined EFDC-LSTM model achieved high-precision forecasts of HABs, offering a reliable approach for the early detection of HABs. On the other hand, the need for more advanced deep learning architectures is increasingly evident for capturing the complex and nonlinear nature of bloom formation. Recent breakthroughs have shown that they can significantly enhance the accuracy of long-term time series prediction tasks in industrial and environmental systems (Bahrainian et al., 2021; Molares-Ulloa et al., 2023; Yang et al., 2022). These ensemble approaches overcome the limitations of individual models by combining their complementary strengths through techniques such as stacked generalization and weighted averaging. The theoretical foundation of multi-model integration lies in the bias-variance tradeoff, where diverse algorithms with different inductive biases collectively reduce prediction variance while maintaining low bias (Zhang et al., 2025).

There has been substantial progress in integrating multiple ML models to improve HABs prediction. Barzegar et al. (2020) proposed a hybrid model that integrates a CNN with LSTM to estimate Chl-a concentrations in a lake in Greece. The input variables were used with lag times up to one (t - 1) and two (t - 2), respectively, to predict Chl-a concentration. The results showed that the hybrid CNN-LSTM model outperformed the independent models (LSTM, CNN, SVM, and Decision Tree (DT) models) in predicting Chl-a. The hybrid model successfully captured both low and high levels of water quality variables. Li et al. (2021) combined the ARIMA model with LSTM to more effectively capture the temporal patterns of highly dynamic observations. The model achieved a significant reduction in predictive errors, with mean absolute error and root mean square error decreased by 12.4 % and 15.5 %. Further advancing hybrid modeling, Ahn et al. (2023) introduced an artificial intelligence deep learning technique designed to directly predict harmful cyanobacteria cell counts. This approach integrates CNN with Transformer algorithms to leverage both models' strengths, enabling simultaneous learning of local and global sequence data patterns. Advanced analyses were performed to evaluate the model's effectiveness, with comparisons drawn to the Temporal Fusion

Transformer (TFT) method. Together, these studies highlight the effectiveness of hybrid and deep learning models in addressing the complex dynamics of algal blooms.

4. Key challenges and future directions for algal bloom identification and prediction

Despite recent advances in artificial intelligence, remote sensing, and ecological modeling, the prediction of inland algal blooms still faces considerable challenges in translating theoretical progress into operational reliability. These obstacles primarily arise from limitations in data availability, model generalization, ecological interpretability, and environmental uncertainties.

4.1. Limited data availability and quality

High-quality data form the cornerstone of any reliable predictive system, yet current observation systems still fall short in both spatial and temporal dimensions. Satellite-based monitoring is often hindered by cloud cover, limited spectral resolution, and coarse revisit times, while in situ stations are sparse and geographically uneven. Most monitoring programs focus on a narrow set of basic physicochemical parameters, such as temperature, pH, and dissolved oxygen, and rarely capture key variables such as nutrient dynamics, underwater light regimes, or phytoplankton community structures. Historical datasets often suffer from inconsistencies, high missing rates, and lack of interoperability, all of which severely restrict model training and long-term validation.

Extensive research on algal bloom forecasting has established that phytoplankton biomass dynamics exhibit significant temporal autocorrelation patterns, demonstrating that bloom development fundamentally operates as a nonlinear autoregressive process. This characteristic necessitates high-frequency monitoring of key indicators such as Chl-a and cell abundance to capture the requisite temporal resolution for effective predictive modeling. Despite this established requirement, conventional water quality monitoring programs frequently implement sampling regimes with insufficient temporal resolution to adequately characterize these autocorrelation structures, substantially limiting the development and operational implementation of reliable forecasting frameworks.

4.2. Poor model generalization and transferability

A major bottleneck in current bloom prediction lies in the limited ability of models to generalize across spatial and ecological contexts. Most algorithms are trained on data from specific water bodies, and often fail to maintain accuracy when applied elsewhere. This regional dependency is especially pronounced in DD models, which are highly sensitive to training data and prone to overfitting. These models heavily depend on patterns within training datasets rather than incorporating fundamental physical, chemical, or biological principles. This dependency makes them particularly vulnerable when applied to novel environments outside their training distribution. The overfitting tendency, especially pronounced in parameter-rich deep learning architectures, causes models to memorize specific "fingerprints" of training water bodies instead of capturing true causal relationships. The inherent complexity of aquatic ecosystems—influenced by numerous nonlinear factors including seasonal variations, climate conditions, nutrient concentrations, and hydrological characteristics—further complicates transferability between regions. Domain shift problems arise when prediction environments differ systematically from training environments in their fundamental parameters like temperature ranges, light conditions, or trophic states. Additionally, the absence of physical constraints or ecological principles as boundary conditions limits these models' inferential capabilities in new contexts. These combined factors result in poor generalization across different regions and ecosystems, substantially reducing the practical utility of DD models in large-scale bloom prediction and early warning systems.

4.3. Lack of ecological interpretability and hybrid modeling

While data-driven approaches such as neural networks can achieve impressive predictive accuracy, they often lack interpretability and provide little insight into the ecological processes driving bloom formation. In contrast, PB models offer mechanistic clarity but tend to be overly complex, parameter-heavy, and difficult to calibrate. Currently, there is a notable absence of hybrid modeling frameworks capable of reconciling predictive power with ecological understanding. Additionally, the failure to incorporate uncertainty analysis into most models reduces their practical relevance for management and policy decision-making.

Recently, various DD models for algal bloom forecasting have experienced explosive growth (Cruz et al., 2021). At the same time, most data-driven models are black box models, with a general trend of decreased interpretability as accuracy increases (Gilpin et al., 2018; Lipton, 2018). XAI research has emerged to improve the transparency of AI systems. XAI encompasses both model-agnostic ex post facto analysis methods, such as Shapley Additive Explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME), and partial dependency plots (PDPs) (Lundberg and Lee, 2017; Salih et al., 2025; Sun et al., 2024), as well as intrinsically interpretable model designs, such as knowledge-guided modeling, physics-informed neural networks (PINNs) (Cuomo et al., 2022; Raissi et al., 2019), and causal-informed architectures. These methods aim to improve model transparency, enhance stakeholder trust, and support hypothesis generation.

However, the application of XAI in the field of HAB prediction remains limited. The highly nonlinear, spatiotemporal, and multi-driver characteristics of HABs make it challenging to obtain stable and generalizable explanations. Ex post facto explanation tools often have difficulty capturing ecological interactions or adapting to baseline changes caused by climate change. Therefore, incorporating domain knowledge into the model structure (rather than relying solely on output-level explanations) may provide a more robust path to achieve interpretable and actionable HAB predictions. The complexity of algal blooms determines that it is difficult for any model to accurately simulate the occurrence of algal blooms. Strengthening the analysis and screening of factors affecting algal blooms will help further improve the model's processing efficiency and prediction accuracy for complex inputs, and it remains one of the research focuses for future applications.

4.4. Amplified uncertainty from external disturbances

The intensification of climate change and human interventions introduces new layers of unpredictability into bloom dynamics. Extreme weather events—such as heatwaves, heavy rainfall, and drought—can rapidly alter nutrient influxes and water column stability, triggering unexpected bloom events. Similarly, anthropogenic disturbances like agricultural runoff, reservoir dredging, and flow regulation introduce episodic and often unquantifiable variability. These factors create feedback and nonlinearities that are difficult to anticipate with conventional modeling tools, further complicating reliable forecasting under real-world conditions.

5. Future opportunities and research roadmap

Moving beyond these challenges will require a paradigm shift toward integrative modeling approaches, data synergy, and actionable insights that bridge scientific knowledge and water management practice. Here, we outline a roadmap for advancing algal bloom prediction through targeted innovations in four strategic areas. Fig. 4 presents a comprehensive strategic framework for machine learning-based HAB prediction and early warning systems. The framework integrates four key components in a systematic workflow. First, multi-source data acquisition

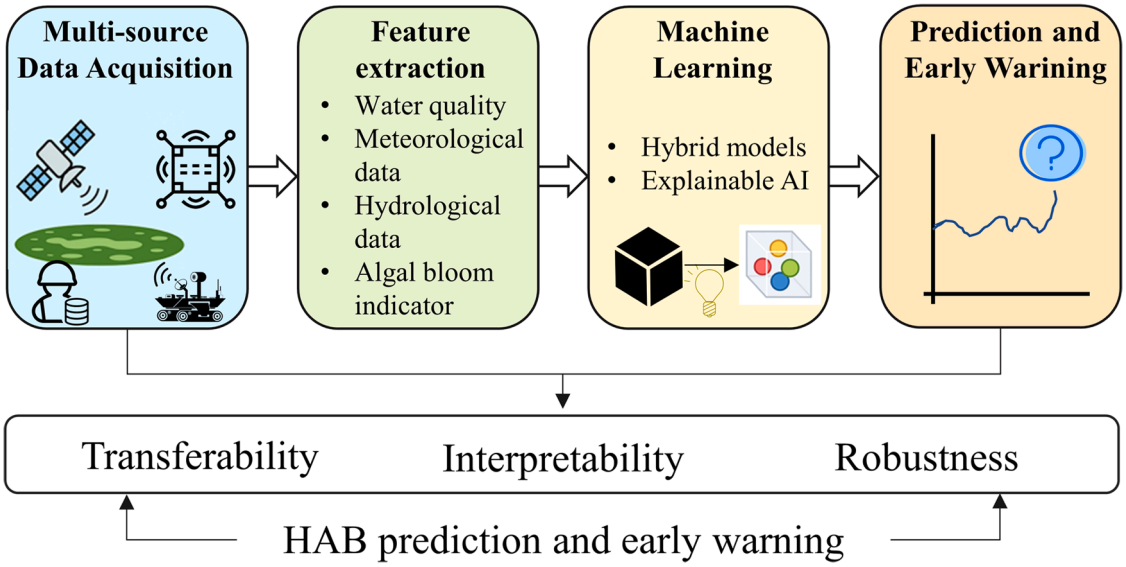

Fig. 4. Strategic framework for machine learning research in HAB prediction.

combines satellite remote sensing for water quality monitoring, meteorological data for environmental conditions, hydrological data for water dynamics, and algal bloom indicators from various sensors and platforms. Second, feature extraction processes raw data to derive relevant variables including water quality parameters (turbidity, temperature), meteorological factors (wind speed, precipitation, solar radiation), hydrological characteristics (flow rates, water depth), and specific algal bloom indicators. Third, the machine learning component employs hybrid models that combine multiple algorithms with explainable AI techniques to ensure model transparency and interpretability. Finally, the prediction and early warning system generates forecasts of HAB occurrence, intensity, and spatial distribution to support timely management responses.

The framework emphasizes three critical qualities essential for effective HAB prediction systems: transferability across different water bodies and geographical regions, interpretability to facilitate scientific understanding and stakeholder acceptance, and robustness to maintain reliable performance under varying environmental conditions.

5.1. Multi-source data fusion for comprehensive monitoring

Improving prediction performance starts with establishing a holistic monitoring framework that combines satellite observations, real-time sensor networks, and manual sampling. Incorporating ancillary datasets—such as meteorological drivers, land use patterns, hydrological data, and anthropogenic inputs—can enrich the feature space and better represent bloom-driving conditions. The use of advanced data assimilation techniques, historical data reconstruction, and quality control protocols will further enhance the robustness and temporal continuity of training datasets. In the future, remote sensing prediction models will be listed alongside process mechanism models and data-driven models as the third type of model for algal bloom prediction. Remote sensing technology extracts water quality information through photothermal sensors carried by satellites and aircraft. It is a means of data monitoring and acquisition and can obtain multi-site, high-frequency remote sensing data. In addition, different algae species contain different pigment contents and types, and have different absorption and reflection spectra of light, showing different colors, which provides a basis for remote sensing methods to monitor specific algae blooms. However, when sensor measurements are inaccurate or inconsistent, training data may contain noise and bias, resulting in poor performance of machine learning models (Li et al., 2020). In order to accurately integrate and fuse information from different sources, sensors need to be calibrated to ensure that data from different sources are consistent and compatible (Mozo et al., 2022). In addition to enhancing the performance of machine learning algorithms, integrated data can also be used to more fully and accurately characterize algal blooms.

5.2. Scalable models through transfer learning and multi-scale integration

To overcome the regional rigidity of current models, future research should embrace machine learning techniques designed for generalization under limited data. Transfer learning and meta-learning frameworks can effectively repurpose models across different lakes or climatic zones, reducing the dependency on site-specific data. Additionally, integrating models across spatial scales—linking high-resolution local dynamics with basin-wide conditions—can help capture context-dependent responses and improve system-wide prediction accuracy.

5.3. Interpretable artificial intelligence for improved interpretability

Bridging the gap between predictive accuracy and ecological interpretability is essential for the practical implementation of HAB forecasting models. Within the broader scope of XAI, two complementary strategies have emerged: (1) embedding domain knowledge directly into model architecture, commonly referred to as knowledge-guided deep learning (KGDL), and (2) applying post hoc explanation tools to trained models (e.g., SHAP, LIME, PDPs).

KGDL approaches incorporate physical, ecological, or hydrological constraints into the structure or training process of machine learning models. For instance, PINNs and ecological regularization methods align predictions with known system behaviors, improving both generalization and interpretability. These methods are particularly promising for forecasting HABs in environments where data is sparse, but domain knowledge is rich.

On the other hand, post hoc XAI tools help reveal how black-box models utilize input features to arrive at predictions. SHAP provides feature attribution at both global and local levels, while LIME approximates decision boundaries using locally interpretable models. PDPs visualize how changes in a single feature affect the predicted outcome. These tools are increasingly used in environmental science to support model validation and stakeholder engagement. To advance HAB forecasting, a multi-tiered approach that combines knowledge-guided model design with post hoc interpretability diagnostics offers a promising

pathway toward models that are not only accurate but also transparent and trustworthy. Such integration aligns with the broader goal of building actionable AI systems for environmental decision-making.

5.4. Quantifying uncertainty and supporting intelligent decision-making

As bloom forecasts become integrated into early warning systems, it is essential to move beyond deterministic predictions toward probabilistic outputs. Uncertainty quantification, sensitivity analyses, and ensemble modeling can provide decision-makers with ranges of expected outcomes, enabling more adaptive and risk-informed interventions. Embedding models into real-time monitoring platforms—supported by IoT infrastructure and cloud-based computing—can facilitate rapid response to bloom outbreaks and support ecosystem resilience at multiple governance levels.

6. Conclusion

Inland algal blooms present a pressing and multifaceted environmental challenge, with profound implications for aquatic ecosystem health, water resource management, and public safety. This review has synthesized recent progress in bloom prediction methods, highlighting the growing integration of remote sensing technologies, ecological modeling, and machine learning techniques—particularly SVM, ANN, and LSTM frameworks. Despite these advances, critical challenges persist in terms of data availability, model transferability, ecological interpretability, and external system uncertainties.

To advance the field, future efforts must prioritize the development of multi-source data fusion systems, hybrid modeling approaches that embed ecological knowledge, and scalable learning frameworks capable of cross-regional generalization. The integration of uncertainty quantification and intelligent decision-support tools will further enhance the operational utility of predictive models, enabling real-time forecasting and adaptive management of bloom events.

Ultimately, addressing the inherent complexity of algal bloom dynamics requires a shift toward transdisciplinary collaboration, coupling innovations in artificial intelligence with insights from limnology, climatology, and hydrology. Such an integrative perspective is essential for building resilient water governance systems that can anticipate and respond to the increasing ecological variability of inland aquatic environments under global change.

CRediT authorship contribution statement

Wanting Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Guoqiang Wang:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Jie Li:** Writing – review & editing, Investigation. **Jinyue Chen:** Investigation. **Zhenyu Gao:** Conceptualization. **Lei Fang:** Conceptualization. **Shilong Ren:** Investigation. **Qiao Wang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article

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